GoServer: a working n-player Go experiment

LeanFM project — playable-v1

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What exists now

The project contains a playable browser board and a Go HTTP server, not only a design proposal. The default local experiment is four colors on an 11 by 11 board. Humans and client-side MCTS can play; history supports sibling variations, and a real animated GIF ends at the selected position. Two independent browser seats can also play one authoritative server game.

This is an implementation experiment explicitly requested by the operator. It does not retroactively accept the contradictory earlier requirements or implementation drafts. The checked specification, runtime tests and observed traffic are empirical evidence, not a proof that arbitrary deployed Go or JavaScript code refines the specification. MCTS optimality and strong multiplayer strategy are not established.

Resume without chat history

This directory is the exclusive project root. Read this book, then `spec/working-contract.json`, `spec/Requirements.lean`, `spec/Requirements.proto`, and `spec/Implementation.lean`. Executable source is under `source/`; tests and evidence are under `tests/` and `evidence/playable-v1/`. The old two-player scaffold under `spec/internal/`, its handwritten protobuf approximation, and the archived `spec/legacy/` drafts are not used by this implementation.

From the project root, `make run` builds and serves the app at `http://127.0.0.1:8082`. No provider key, paid model call or external Go dependency is involved. `make check`, `make browser-check` and `make book` reproduce the checks and PDFs. Dependencies are listed in `README.md`. Runtime state remains in `.runtime/`; test caches, browser profiles and complete compiler logs remain in `.cache/`.

Exact game rules

Open is an empty playable intersection. A group is a maximal orthogonally connected set of one player's stones. A liberty is an adjacent Open point. The implicit exterior Wall is unplayable and uncapturable; it is *not* a liberty for player stones. This corrects the previous book's accidental immortal-edge-group rule.

A move places the current player's stone on Open. Collect every adjacent opposing group with zero liberties and remove those groups simultaneously. If no opponent was removed, remove the mover's own group when it has no liberties. Suicide is allowed. Each removed stone increases its owner's cumulative losses. In the capture matrix, a row is the capturer and a column the removed color: other-color captures are positive; own-color removals are negative diagonal entries. Captures give no separate score bonus.

An Open region is territory for a color precisely when its non-Wall boundary contains that one color and no other. A region with no player boundary or a mixed boundary is neutral. Score is territory

minus losses. Repeated board positions are allowed; cumulative losses do not reset. No ko prohibition is silently added to this experiment.

Pass is always available. Resign removes that player from turn rotation. The game ends after a consecutive pass by every active player, one active player remains, or the configured turn limit is reached (default 200). Each live turn has a 60-second deadline. The browser observes its local deadline; the server catches up overdue Pass actions on the next state read or action. An offline room therefore need not have a continuously running timer thread. No automatic placement is invented by the server.

Interaction and authority

Local mode uses immutable JavaScript positions and a branching analysis tree. Arrow keys navigate the labeled grid; Enter or Space plays. Back and Forward select history; a move from an earlier local node preserves sibling variations. Losses, scores and the signed matrix are ordinary labeled HTML tables, not information available only inside the canvas.

MCTS runs in a cancellable worker, evaluates a rank utility for every player, and includes Pass. The selectable 0.1–8 second thinking budget fits inside the 60-second turn deadline. Rollouts are bounded heuristic evaluations, not guaranteed optimal terminal play. A fixed seed and fixed work budget support reproducible tests; wall-clock search varies with host performance.

In a network room, every placement, Pass or resignation is a proposal. The server checks the seat token, turn, revision and rules, persists the complete updated room, then publishes the new state. History review cannot submit a move. Reload restores only the original tab's session credential; an invitation alone cannot claim an existing seat. Sharing an invitation grants spectator read access and permits joining an unoccupied seat. Rooms have no private-data or public-production deployment claim; the service binds to loopback only.

The wire contract is a requirement

`Requirements.lean` owns six HTTP task contracts: create, join, play, inspect state, inspect history and export. Methods, routes, codecs, statuses, authentication and correlation must not be overridden by the implementation. `Requirements.proto` owns the typed values, including repeated nested positions and signed ledger entries. `http://127.0.0.1:8082/openapi.yaml` serves the mechanically derived OpenAPI document.

JSON responses retain normalized observed events. Every accepted request has one terminal success or error; a successful turn contains MoveEvent before GameState. Responses retain the exact request ID as `cause_id`, and `prior_ids` separately orders the events. GIF responses carry the same event array in the base64 `X-GoServer-Trace` header. Host and Origin admission failures are plain HTTP errors before an accepted task begins.

Failure and security boundaries

R1–R6 in the working contract link to controls and negative tests in `evidence/playable-v1/traceability.json`. Illegal actions, wrong seats, missing credentials, duplicate submissions and stale revisions cannot change the requested position. Failed persistence does not publish an uncommitted action. Snapshots store token hashes, never plaintext seat tokens. Restart replays and compares every saved position before accepting a room.

Host validation rejects DNS rebinding; Origin validation rejects cross-origin mutations. Strict UTF-8 JSON rejects duplicate/unknown fields, excess nesting and oversized bodies. The CLI rejects escaping or symlinked runtime paths. Boards are at most 19 by 19, there are at most six colors and 512 turns,

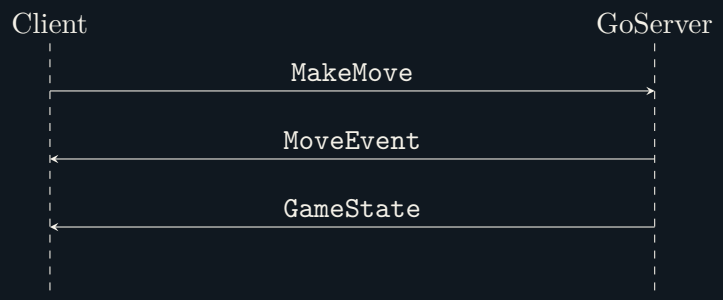


Figure 1: Observed successful turn, generated from the actual protocol test trace, not an independently invented example. Prompt: actually play a game between two users. Question: did this exchange follow the grammar? Source: `evidence/playable-v1/protocol.json`. Reducer: grammar acceptance plus exact cause/prior checks. Unit: events; boundary: one HTTP turn.

local analysis holds at most 2,048 nodes, and storage holds at most 32 rooms and 64 exports. A GIF contains at most 200 actions; export memory is serialized. The intended scope is a trusted local operator, not an Internet-facing service.

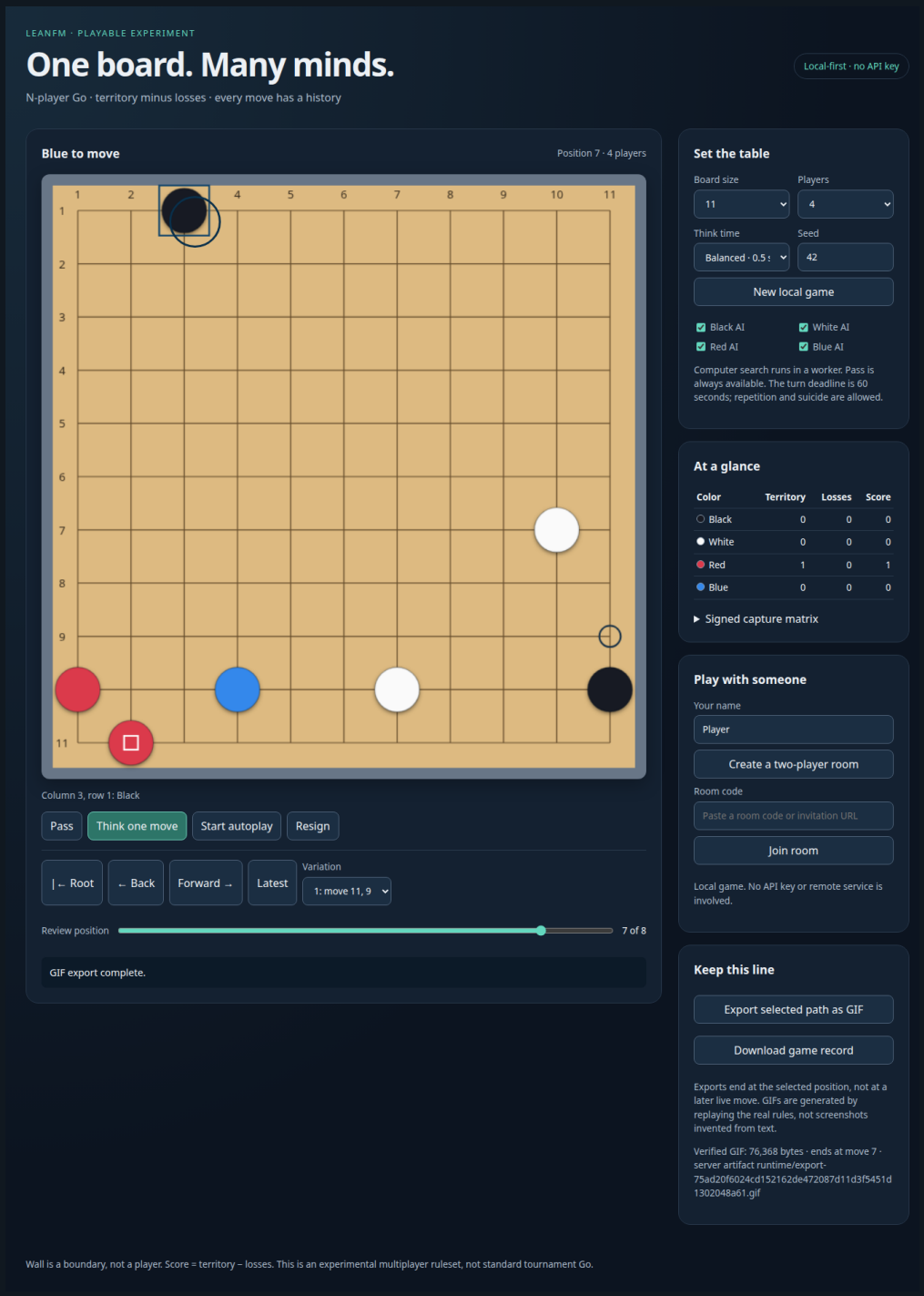
No screenshot is claimed until the real browser produces it. GIF export replays selected actions through Go rules, verifies them, then encodes actual GIF bytes; it never accepts arbitrary filesystem paths. Errors are surfaced with their failing operation. There is no automatic paid repair/retry loop. Both PDFs are compiled in scratch directories and promoted only after both variants succeed; full compiler output is retained on failure.

Verification and limits

The differential suite compares every intermediate board, loss ledger, turn and score in 40 seeded games (2,240 transitions). Go race tests cover independent seats, simultaneous submissions, deadline Passes, rejected input, persistence failure and restart. Real HTTP scenarios for all six tasks round-trip through official protoc-generated bindings and the Lean-derived grammars; reversed and truncated traces are rejected. Browser traces are checked too.

The browser suite exercises MCTS, keyboard play, sibling variations, selected GIF download, two isolated seats completing a game, reload recovery and all 121 accessible grid labels. The screenshot on the next page is a real capture. `board-labels.txt` and `browser.json` retain its labels and selected state. These finite checks do not establish general grammar recognition, checker correctness, strategic competence, network fault tolerance or a formal Go/JavaScript refinement proof. Those remain explicit follow-up work.

Captured running application



Prompt: produce a working n-player Go application. Question: does the actual served UI expose the required controls and selected state? Source: the real headless browser DOM and canvas. Reducer: capture the selected position after MCTS, keyboard play and history review. Unit: pixels and labeled cells; boundary: one browser viewport. This static PDF is not an animation; the real animation is `evidence/playable-v1/selected-line.gif`.